

KORNU SAI GOVINDA RAO

kornusaigovinda@gmail.com | +91 8074025622

[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [LeetCode](#) | [Hackerrank](#)

PROFESSIONAL SUMMARY

Full Stack Developer and Data Scientist with hands-on experience building AI-powered platforms end-to-end from Angular/FastAPI application architecture to LLM integration, prompt engineering, and machine learning model development. Skilled in designing responsive, production-grade frontends, optimizing backend performance through database indexing and Redis caching, and deploying scalable applications on AWS. Experienced in owning projects independently, from a live personal platform (Algonimate) to freelance client delivery, with a strong foundation in data analysis, model evaluation, and cost-efficient AI workflow design.

SKILLS SUMMARY

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Frontend Development: Angular, Ionic, Responsive Web Design, REST API Integration, Dashboard Development, Data Visualization

Backend Development: FastAPI, REST API Design, Authentication & Authorization, Server-Side Development, Backend Workflow Design

Database & Performance: MySQL, Database Design, Query Optimization, Indexing, Redis Caching, Efficient Data Retrieval

AI/ML & Data Science: Machine Learning, Deep Learning, Exploratory Data Analysis, Feature Engineering, Model Evaluation, LLM Integration, Prompt Engineering

ML/Data Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib

Cloud & DevOps: AWS, Docker, Netlify, Git, Bitbucket

Developer Tools: VS Code, PyCharm, Jupyter Notebook, Google Colab, MySQL Workbench

Core Strengths: Analytical Thinking, Problem Solving, Cross-functional Communication, Team Collaboration, Technical Leadership

WORK EXPERIENCE

Data Scientist - SpadeCode Intelligence Pvt. Ltd. | *April 2025 – Present*

- Own end-to-end development of SpadeCode, an AI-powered interview and assessment platform, working across Angular frontend, FastAPI backend, database design, and LLM-based workflows, while guiding team members on implementation and production-grade solutions.
- Design and optimize LLM-driven interview workflows that generate personalized questions, follow-ups, and feedback from user inputs (domain, resume, job description), including prompt engineering and token-cost optimization across mock interviews and tests.
- Built an end-to-end audio interview pipeline integrating Whisper-based speech-to-text, WebSocket communication, and real-time transcription, while analyzing processing costs to optimize the overall AI cost pipeline.
- Developed and evaluated Machine Learning and Deep Learning models using Python, applying industry-standard evaluation metrics and model validation techniques, contributing to broader AI-driven product features beyond the core LLM workflows.
- Engineered database indexing and Redis caching strategies to enable fast retrieval across large question datasets, improving response times for high-traffic mock-test workflows.
- Developed candidate dashboards, performance analytics, and leaderboard features using Angular, consolidating interview results, feedback, and progress trends to improve user engagement.
- Integrated Razorpay payment gateway with location-based currency handling and led AWS deployment operations (EC2, RDS, S3, CloudFront, CloudWatch) for production hosting, monitoring, and troubleshooting.

- Progressed from Intern to Data Scientist, taking ownership of technical planning, sprint coordination, and mentoring team members on scalable, production-oriented engineering practices.

Freelance Full Stack Developer | srrrigservices.in | Jun 2026

- Designed and developed a production-grade business website using Angular, delivering a responsive, user-friendly interface tailored to client requirements and end-user navigation needs.
- Owned the complete website lifecycle end-to-end, including domain configuration, deployment, hosting setup, and production release, ensuring a smooth transition from development to live operation.
- Implemented on-page SEO optimization, including metadata structuring, search-friendly content architecture, and indexing configuration, to improve organic search visibility and discoverability.
- Collaborated directly with the client to gather business requirements, translate them into functional website features, and independently manage the project from planning through deployment.

Personal Project - Algonimate | DSA Visualization Platform | algonimate.com | Sep 2026

- Designed and built Algonimate, a free interactive platform visualizing 40+ data structures and algorithms including sorting, linked lists, stacks/queues, dynamic programming, backtracking, and graph/tree traversal to help learners grasp algorithmic concepts through animation.
- Engineered step-by-step algorithm playback using GSAP and SVG/Canvas rendering, building a reusable animation and state-tracking system to synchronize visual transitions (comparisons, swaps, traversals) with algorithm execution in real time.
- Developed the platform end-to-end using Angular and TypeScript, building a responsive, theme-aware (dark/light) UI with reusable component architecture optimized for smooth animation performance across devices.
- Owned the complete product lifecycle independently domain acquisition, deployment, hosting, SEO optimization, and ongoing maintenance making the platform globally discoverable and accessible to self-taught learners.
- Actively expanding scope with planned Machine Learning workflow visualizations and software-engineering concept modules (caching, API flows, database indexing), extending the platform beyond DSA into broader CS education.

PROJECTS

Drone vs Bird Detection & Classification | [Link](#)

- Built an end-to-end computer vision system using custom CNN and InceptionV3 transfer learning to classify drones vs. birds in aerial imagery, applying preprocessing, augmentation, and dataset balancing for improved generalization.
- Integrated YOLOv8 for real-time object detection and bounding box localization, achieving ~96% classification accuracy with a deployable prediction pipeline for unseen images.

AI-Generated vs Real Image Detection | [Link](#)

- Developed a deep learning system using custom CNN and EfficientNetB0 transfer learning to distinguish AI-generated images from authentic photographs, applying extensive preprocessing and augmentation for robust performance.
- Built a real-time prediction pipeline for user-uploaded images, achieving ~91% accuracy and validating the effectiveness of pretrained models for authenticity verification.

Loan Approval Prediction System | [Link](#)

- Designed a machine learning system to automate loan approval decisions, performing extensive EDA, feature engineering, and class-imbalance handling across multiple models (Logistic Regression, Random Forest, Gradient Boosting).
- Achieved ROC-AUC up to 0.975 with optimized recall to minimize financial risk, and deployed an interactive Streamlit application for real-time prediction.

EDUCATION

MCA (Master of Computer Applications)	Aug-2024
Dr. Lankapalli Bullayya College	CGPA: 8.4
Bachelors of Science in Mathematics	Oct-2021
Sagar Degree College	CGPA: 7.5
Intermediate, MPC	Apr-2018
Sri Venkata Sai Jr College	Percent: 85.4

CERTIFICATIONS

Python Basics	:	By Hacker Rank	NumPy	:	By Great Learning
SQL Basics	:	By Great Learning	MS-Office	:	By Samavedha It Solutions